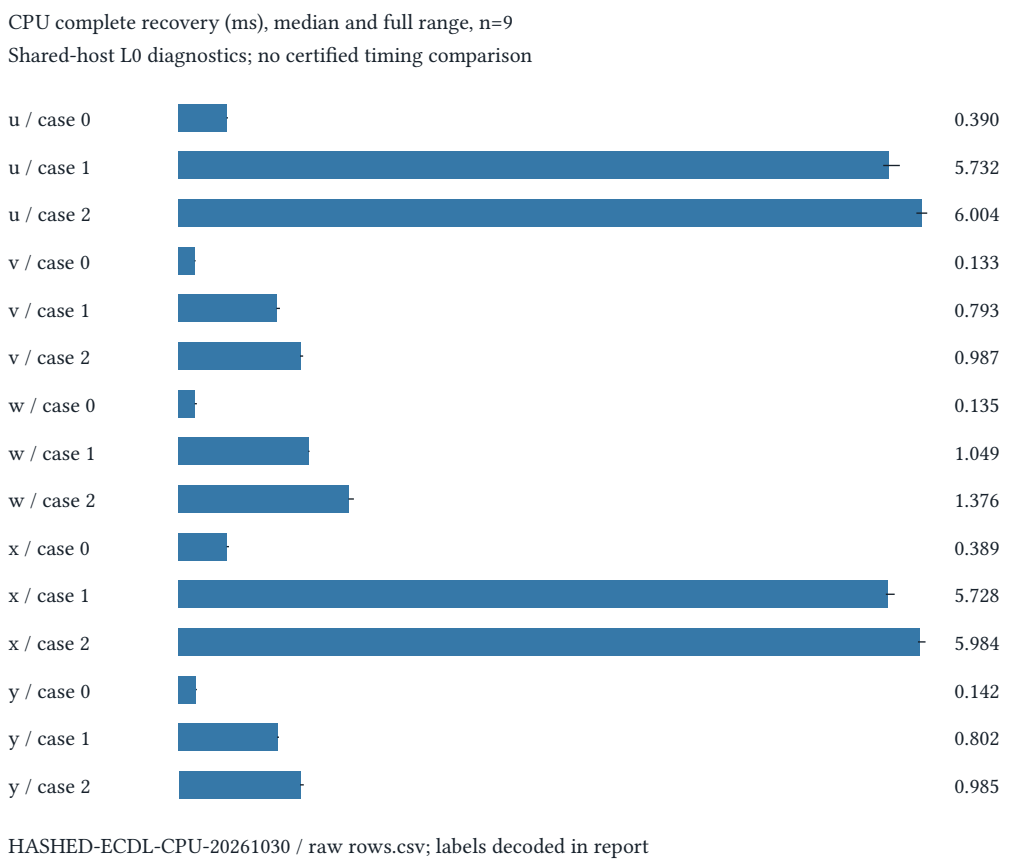
CPU bounded field-inverse cache

Shared-host observations only; no certified speedup or hypothesis promotion. Job `hashed-ecdl-cpu-round16-20261030`, October 30, 2026 UTC.

Integrated source: candidate v/w `5d7dcb67616c`, pre-cache reference y `4c03472f6bdd`, original baseline u/x `2bd4588fba60`. Intel Xeon 6975P-C, one software thread, 2 vCPUs / 4 GiB reserved. No cross-panel pooling.

Fixture	Baseline u ms	Pre-cache y ms	Candidate v ms	y/v
8-bit / 3 slots	0.390351	0.142364	0.132843	1.072
16-bit / 3 slots	5.732379	0.802128	0.793350	1.011
16-bit / 4 slots	6.003614	0.984624	0.986796	0.998

Medians from nine measured repetitions after one warmup. Baseline u/v ratios: 2.938, 7.226, 6.084. Ratios are derived from measured medians. The further 10x target remains unmet.



The candidate builds field inverses for validated primes p <= 4096 and uses them in affine additions, projective normalization and residual batch inversion. Each Curve owns its table; construction remains charged to setup. Larger fields retain extended Euclid. Added memory is 8*p bytes plus Vec overhead, at most 32 KiB of entries; the retained sum-table metric excludes it. Inversion counters count requests, including cache lookups, not Euclid calls.

All 150 executions completed (135 measured,15 warmups). Full certificate bytes, scalar checks, replay and corrupted-scalar rejection pass. Setup, discarded trials, elimination, both checks and serialization remain charged. Launch, archive I/O and negative controls are excluded; external timers are unchanged.

Candidate v ranges (ms): 0.129832–0.137313,0.789557–0.817929, 0.980129–1.005459; reference y: 0.140528–0.147706,0.795359–0.810315, 0.980676–1.010528. The smallest fixture's ranges do not overlap; the two larger ranges overlap. A/A x medians: 0.389038,5.727942,5.984068ms. No certified timing improvement or equivalence follows from this L0 panel.

The34-test candidate passes strict Clippy and formatting. The export passes completeness and file-integrity checks with zero findings. Returned build, protocol and both source manifests match the frozen files. Full root/CUDA build, strict isolation, independent review, external/rho comparisons and larger-scale CPU recovery remain outstanding. Protocol: `33e294d07d0ffc6f66fb696eac19d01a00c14add29e4532e48ea5e78c3d580c9`. Raw rows, certificates and checked export: `round16/results/`.